

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

CHEMICAL ENGINEERING
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Which quantity expressed with its magnitude and unit?	M 1.01	R
2	What is the definition of gm atom?	M 1.03	R
3	The ability of a substance to conduct heat is called _____	M 2.02	R
4	List two evaporators used in industry.	M 2.04	R
5	The term used for expressing the efficiency of equipment's in mass transfer operations is _____	M 3.01	R
6	In a mass transfer process , as temperature increases, solubility _____ (a) Increases (b) Decreases (c) Remains constant (d) None of the above	M 3.02	R
7	The temperature at which the saturated vapour starts to condense is called _____	M 4.02	R
8	Constant boiling mixtures is also known as _____	M 4.04	R
9	Define the term diffusivity.	M 3.01	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	What differences exists between fundamental and derived units?	M 1.01	U
2	Estimate the heat loss per m ² of the surface through a brick wall 0.5m thick when the inner surface is at 400K and the outside surface is at 310 K. The thermal conductivity of the brick may be taken as 0.7 W/(m.K)	M 2.02	A
3	Explain the terms raffinate and extract in liquid-liquid extraction?	M 3.02	U
4	Illustrate steam distillation.	M 4.01	U
5	State Raoul's law.	M 4.02	U
6	State Fick's law of diffusion.	M 3.02	U
7	Explain simple distillation with a neat sketch.	M 4.01	U

8	A wall is made of brick of thermal conductivity 1.0 W/ (m.K), 230 mm thick. It is lined with plaster of thermal conductivity 0.4 W/(m.K) and of thickness 10 mm. If a temperature difference of 30K is maintained between the two faces, determine the heat flow per unit area of wall.	M 2.02	A
9	What is continuous leaching?	M 3.02	U
10	Write short notes on extractive distillation.	M 4.04	U

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain various systems of units	M 1.01	U
OR			
2	A solution prepared by dissolving 66.0 g of urea (NH ₂) ₂ CO in 950 g of water had a density of 1.018 g mL ⁻¹ . Express the concentration of urea in a) weight- per cent; b) mole fraction; c) molarity; d) molality.	M 1.03	A
3	Explain the working of Shell and Tube Heat Exchanger with a neat sketch.	M 2.03	U
OR			
4	Explain the working of horizontal tube evaporator.	M 2.04	U
5	Explain the working of Packed Tower.	M 3.01	U
OR			
6	Explain the working of a spray drier.	M 3.04	U
7	Compare distillation with extraction.	4.01	U
OR			
8	Describe maximum and minimum boiling azeotropes.	4.04	U
9	Explain the working of double pipe heat exchanger with a neat sketch.	M 2.03	U
OR			
10	Derive heat transfer equation by conduction through plane wall.	M 2.02	U
11	Explain density and specific gravity. Calculate the density in g/mL of 3000 mL of solution weighing 6 kg.	M 1.04	U
OR			
12	Differentiate Normality from Molarity. Calculate the normality of HCl if 21.18 mL is used to titrate 0.1369 g Na ₂ CO ₃ ?	M 1.03	A

FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

CHEMICAL ENGINEERING
MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A

IV. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Define the term normality.	M 1.04	R
2	List two examples for unit operations.	M 1.02	R
3	The fastest mode of heat transfer is through _____	M 2.01	R
4	List any two heat exchanger patterns used in an industry.	M2.03	R
5	The efficiency of the equipment in a mass transfer operation is measured in _____	M 3.01	R
6	Which process is used in metals industry for removing mineral from ores?	M 3.03	R
7	Define Ideal solution.	M 4.02	R
8	The temperature at which the liquid starts to boil is called _____	M 4.01	R
9	Name the separation technique used in petroleum refineries and chemical plants.	M 4.03	R

PART B

V. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	Calculate the molarity of 25.0 grams of KBr (molecular weight = 119 g/mol) dissolved in 750.0 mL.	M 1.04	U
2	Write a short note on the mechanism of natural convection.	M 2.01	U
3	What is flooding and channelling in packed columns?	M 3.01	U
4	List the characteristics of minimum boiling azeotrope.	M 4.04	U
5	Write a short note on liquid-liquid extraction.	M 3.03	U

6	What are the characteristics of non-ideal solutions?	M 4.02	U
7	Write a short note on vapour liquid equilibria for a binary mixture	M 4.01	U
8	A steam pipeline, 150/160 mm in diameter, carries steam. The pipeline is wrapped with a layer of insulating material (thermal conductivity 0.08 W/ (m.K)) of thickness 100 mm. The temperature drops from 392.8 K to 313 K across the insulating surface. Determine the rate of heat loss per 1 m length of pipe line.	M 2.02	A
9	Illustrate the working of tray drier.	M 3.04	U
10	Calculate the equilibrium composition of the liquid and vapour phases for a mixture of methyl alcohol and water at a temperature of 323 K and under a pressure of 40 KPa. Assume that both the liquid and vapour behave ideally. Vapour pressure of methanol and water at 323 K are 53.32 KPa and 12.33 KPa respectively.	M 4.02	U

PART C

VI. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Calculate the mole fraction of ethanol and water in a sample of rectified spirit which contains 95% of ethanol by mass	M 1.02	A
OR			
2	Explain different methods of expressing concentrations of solutions.	M 1.03	U
3	Derive the equation to calculate heat transfer through cylinder.	M 2.01	U
OR			
4	A furnace is constructed with 225 mm thick of fire brick, 120 mm of insulating brick and 225 mm of the building brick. The inside temperature is 1200 K and the outside temperature is 330K. Find the heat loss per unit area and the temperature at the junction of the fire brick and insulating brick. Data: k for fire brick, insulating brick and building brick are 1.4, 0.2 and 0.7 W/ (m.K) respectively.	M 2.04	A
5	Illustrate the extraction of cotton seed oil using Boll man extractor.	M 3.02	U
OR			
6	Describe the construction details and working of Mixer settler.	M 3.03	U
7	Explain Raoult's law and derive its equation	M 4.02	U

OR			
8	Define relative volatility and derive the equation for relative volatility.	M 4.03	U
9	Derive the equation to calculate heat transfer through composite plane wall.	M 2.03	U
OR			
10	Explain the working of multiple effect evaporators.	M 2.04	U
11	Explain the concept of unit operation and unit process.	M 1.02	U
OR			
12	4.6cm³ of methyl alcohol is dissolved in 25.2g of water. Calculate (i) percentage by mass of methyl alcohol (ii) mole fraction of methyl alcohol and water. Given density of methyl alcohol = 0.7925g/cm³	M 1.03	A